Evaluating the Granularity Balance of Hierarchical Relationships in Large Biomedical Ontologies

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Abstract

Organizing the descendants of a concept under a particular semantic relationship may be rather arbitrarily carried out during the manual creation processes of biomedical ontologies, resulting in imbalances in relationship granularity. In this paper, we utilize "*parallel concepts set*" and structural information to systematically evaluate the granularity balance of semantic relationships within large biomedical ontologies. The criteria for balanced relationship granularity is proposed: The different traces along a particular semantic relationship from one set of parallel concepts to another are expected to be balanced in terms of "*semantic distance*," which is calculated based on the length and the strength of each trace, represented by the number of steps and the aggregation of semantic weights, respectively. General evaluation models are designed based on the above criteria, and experiments are conducted on two kinds of parallel concepts sets : one is with a single concept and the other one contains a pair of symmetrical concepts. We checked the granularity balances for the IS-A and Part-Of relationships in FMA and SNOMED CT's Body Structure subsets. Results show that there are 121 instances of imbalances in FMA and 124 of that in SNOMED CT for the IS-A relationship. Moreover, 6801 instances of imbalances for the Part-Of relationship in FMA were also detected, including 4935 length imbalances and 1866 strength imbalances.

Keywords: Ontology Quality Assurance, Biomedical Ontology, Granularity Balance, Parallel Concepts

1. Introduction

Biomedical ontologies and controlled terminologies play a vital role in the field of biomedical informatics [1, 2, 3], which further demonstrates the importance of Ontology Quality Assurance(OQA) [4, 5, 6, 7]. As the key component of biomedical terminologies, semantic relationships inevitably became the hot spot of auditing methods, varying from lexical [8, 9], structural [10, 11, 12, 13, 14, 15, 16], semantical [17, 18, 19], statistical [20], to abstraction-network based [21] and big data based [22, 23].

For a particular semantic relationship, how to select and organize the descendants of a concept is a challenging problem. The ideal scenario is that the semantic distances between every pair of directly related

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concepts(i.e.,Parent and Child)are identical. Same for the distances among siblings. However in reality, although domain expert knowledge was involved during the creation processes, inconsistencies and imbalances in relationship granularity may still be introduced into biomedical ontologies. By "granularity of a relationship r" in this work we mean the level of detail at which knowledge is presented in a biomedical ontology [24, 25], demonstrated by the semantic distance between two concepts directly related by r. The longer the distance, the coarser the granularity, and vice versa.

Imbalanced granularity of relationships turned to be a barrier for ontology mapping, semantic interoperability and data integration. However, auditing methods on the granularity of relationships were quite few in the literature. Sun and Zhang [26] identified the granularity differences between two biomedical ontologies through rules to distinguish among different types of subclasses and classifications. He et al. [27, 28] adopted the term "density" instead of granularity and defined a set of topological patterns to demonstrate different concept densities across pairs of terminologies, aiming to enrich SNOMED CT conceptual content and support semantic harmonization of SNOMED CT with other Unified Medical Language System (UMLS) terminologies.

These past works have focused on granularity differences across terminologies using comparative study, while we propose a novel systematic mechanism for auditing the intrinsic granularity balance of relationships within a targeted biomedical ontology in this paper. Currently, there is no method for calculating the granularity of semantic relationships directly. Therefore, we choose to examine the granularity balance relatively by utilizing "*parallel concepts set*." Two concepts are parallel if they share similar level of conceptual knowledge, such as symmetrical concepts in FMA [10]. As a result, a parallel concepts set contains a number of concepts who are parallel to each other. The assumption is: If the granularity of a relationship in an ontology is balanced, then all the traces along this relationship from one set of parallel concepts to another one are expected to be balanced in terms of "*semantic distance*," which is represented by the length and the strength of each trace.

The length of a trace is the number of steps, and the strength is introduced to deal with cases when the semantic relationship has a hierarchy of subproperties, for example, the Part-Of relationship in FMA has a hierarchy of three levels, as shown in Figure 1. Subproperties of each hierarchy are assigned a distinct semantic weight and the strength of a trace is the aggregation of all the semantic weights of the subproperties along it. The granularity is balanced if the lengths of all the traces are the same, as well as the strengths, if needed. Figure 2 demonstrates two examples with imbalanced length and imbalanced strength in SNOMED CT and FMA, respectively.

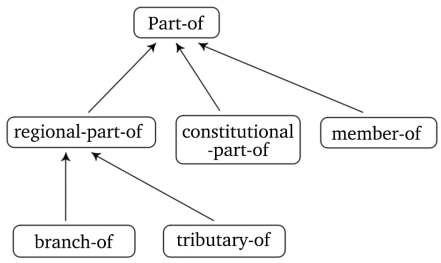

Figure 1: The three-level hierarchy of the Part-Of relationship in FMA

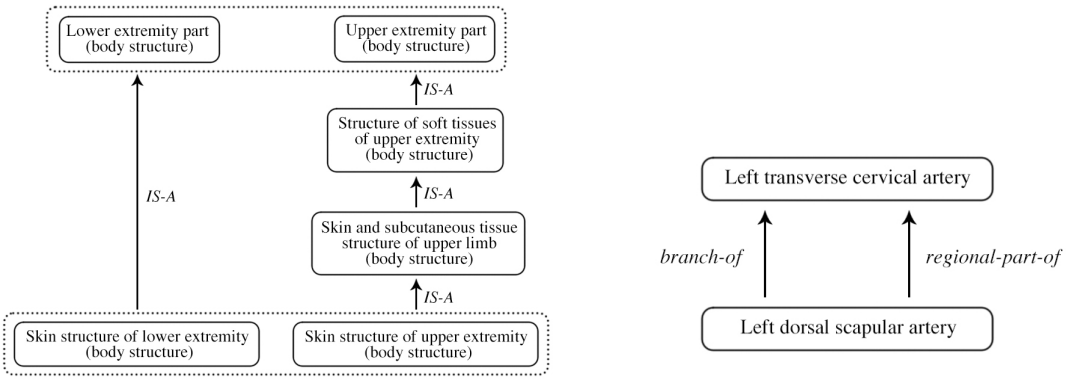

Figure 2: Left: An example with imbalanced length in SNOMED CT; Right: An example with imbalanced strength in FMA

Based on the idea above, taking the length and the strength aspects into account, we first propose two general models for evaluating the granularity balance of relationships. Then, we select two kinds of parallel concepts sets for experiments: one contains a single concept in the set, as illustrated by the example on the right of Figure 2, and the other one contains a pair of symmetrical concepts, as illustrated by the example on the left of Figure 2. As demonstrated in our previous work [10], symmetrical concepts are bisimilar and suffice to be parallel concepts. Based on the two types of parallel concepts sets, we achieved four groups of specialized evaluation models. At last, we applied these models to the IS-A and Part-Of relationships in FMA and SNOMED CT's Body Structure subsets. Results show that there are 121 instances of imbalances in FMA and 124 of that in SNOMED CT for the IS-A relationship. On the other hand, 6801 instances of imbalances for the Part-Of relationship are also detected, including 4935 length imbalances and 1866 strength imbalances.

In all, this study presents a novel systematic mechanism to detect intrinsic granularity imbalances of semantic relationships within large biomedical ontologies for quality assurance. The criteria for balanced granularity of relationships is proposed for the first time, and the corresponding evaluation models are constructed, which are proven to be effective in uncovering instances of imbalances by experiment results.

2. Background

2.1. The Foundational Model of Anatomy (FMA)

The FMA is both a theory of human anatomy and an ontology artifact [29]. As a theory, it provides a unifying framework for the nature of the diverse entities that make up the bodily structure of biological organisms as well as the relationships between them. In particular, it is a theory of the canonical, phenotypic structure of the human organism at all biologically salient levels of granularity. As a theory of *canonical anatomy*, it ranges over those categories of entities which are idealizations of an organism's body and its

typical component parts. As a computational artifact, it is a formal representation of this theory, suitable for machine manipulation. The model underlying the FMA is a frame-based representation with more than 80,000 concepts including macroscopic, microscopic and sub-cellular canonical anatomy. For our analysis, we used the version 4.4.0 of FMA created in June, 2016. It is a version distributed as an RDF/XML-based serialization that enables it to be stored in an RDF data store and made available to be queried via SPARQL over internet protocol.

Concepts in ontology are organized into a concept hierarchy with more generic concepts modeled as parents of more specific concepts using the *subClassOf* property (e.g. , Right Hand *subClassOf* Hand). This relationship is called the IS-A relationship. Another important relationship in ontologies is the Part-Of relationship (e.g. Hand *part_of* Free upper limb). In this paper, we focus on auditing these two relationships.

2.2. SNOMED Clinical Terms (SNOMED CT)

SNOMED Clinical Terms (SNOMED CT) is the largest clinical terminology in the world [30, 31], developed and managed by the International Health Terminology Standard Development Organization (IHTSDO) [32]. It provides standard for encoding clinical content, ranging from clinical findings, procedures, to diseases and diagnoses in electronic health records (EHRs), aiming to enhance interoperability. Concepts in SNOMED CT are arranged into 19 subhierarchies including Body Structure. In this study, we used the US Edition of SNOMED CT released in March 2016, and focused on its Body Structure subset, which contains almost 30,000 concepts.

3. Methods

3.1. General Evaluation models

Apparently, it is hard to calculate the degree of granularity of relationships directly. So we work in a relative way and audit the balance of granularity by evaluating the semantic distances between parallel concepts sets. A parallel concepts set contains a number of concepts who share similar level of conceptual knowledge, such as symmetrical concepts in FMA [10]. For two parallel concepts sets $\{P_1, P_2, ..., P_i\}$ and $\{C_1, C_2, ..., C_j\}$ and a semantic relationship r, if there exists more than one r-traces from $\{C_1, C_2, ..., C_j\}$ to $\{P_1, P_2, ..., P_i\}$, the semantic distance of each r-trace is evaluated in two aspects:

- The length of the trace, defined as the number of steps. The lengths of all the r-traces are expected to be identical if r is balanced.
- The strength of the trace in cases that r has a hierarchy of subproperties. Subproperties of r at the same level are assigned a distinct semantic weight, and the strength is defined as the aggregation of all the semantic weights along the trace. The strengths are also expected to be the same among all r-traces.

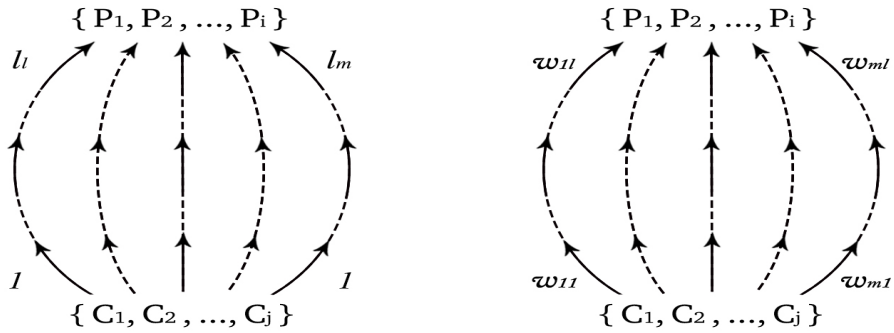

Figure 3: The general models for evaluating semantic distances between two parallel concepts sets. The left $l_1 : l_2 : ... : l_m$ model evaluates the length and the right model evaluates the strength(all the traces have the length l).

For a semantic relationship r, Figure 3 gives the general models for evaluating semantic distances between two parallel concepts sets in terms of length and strength respectively. In the left model, $l_k(1 \leq k \leq m)$ represents the length for all the m r-traces. We denote the model as a $l_1 : l_2 : ... : l_m$ model. If there exists $k_1 \neq k_2$ such that $l_{k_1} \neq l_{k_2}$, which means, the semantic distance is imbalanced in length, as a result, r is determined as imbalanced in granularity. In the right model, suppose there are m traces and the lengths of them are the same, denoted as l, and the semantic weight of each step is denoted as $w_{ij}(1 \leq i \leq m$ and $1 \leq j \leq l)$, then $\Sigma w_{ij}(1 \leq j \leq l)$ represents the strength of the ith trace. The strengths of all the m traces are expected to be the same if relationship r is balanced in granularity.

In the remainder of the paper, we specialize the models in Figure 3 and focus on the $1 : n(n \geq 1)$ pattern instead of the generalised $l_1 : l_2 : ... : l_m$ pattern based on the fact that most of the $m : k(m, k \geq 1)$ patterns are essentially $1 : n$ patterns.

3.2. Specialized Evaluation models

Two types of parallel concepts sets are selected to evaluate the granularity balances of semantic relationships in both FMA and SNOMED CT: 1) Type I: single concept set containing only one concept; 2) Type II: symmetrical concepts set containing a pair of symmetrical concepts. As demonstrated in our previous work [10], symmetrical concepts are bisimilar and suffice to be parallel concepts. As a result, the four combinations of the two types generate four cases of evaluation models, which are demonstrated by the following four Figures(from Figure 4 to Figure 7). In each Figure, there is a $1 : l$ model on the left for evaluating the relationship by length, and on the right, there is a $1 : 1$ model for evaluating the relationship by strength, where r and r' represent two subproperties of the same relationship at different levels, if they exist.

3.2.1. Case (I): From Type I to Type I

The most special parallel concepts set is the set containing one single concept. In Case (I), the source and the destination of the traces in the models are both single concepts. Let the source be concept C and the destination be concept P, the models for evaluating balances of a semantic relationship by length and by strength separately are demonstrated in Figure 4. As discussed before, most of the $m : k(m, k \geq 1)$ cases are essentially $1 : l$ cases, so we only consider the $1 : l$ model for evaluating the relationships by length and the $1 : 1$ model for evaluating the relationships by strength. Note that the labels along the traces in the left graph represent the number of steps, while the labels along the right one represent relations at different levels.

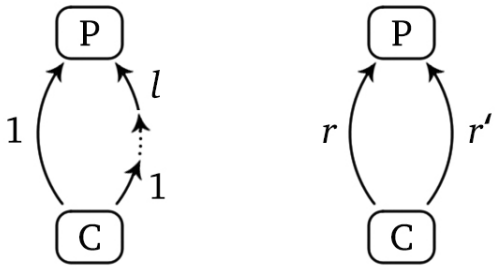

Figure 4: Models of Case (I). On the left is the $1 : l$ length model; On the right is the $1 : 1$ strength model.

3.2.2. Case (II): From Type I to Type II

In Case (II), the source is a single concept while the destination is a set containing two symmetrical concepts. We use pair (u, v) to stand for a pair of symmetric modifiers such as *(left, right)*. Putting the symmetric modifiers into the same context C results in a symmetrical concept pair $(C[u], C[v])$, such as *(left heart, right heart)*. Figure 5 demonstrates the two models with traces from concept A to the set $(C[u], C[v])$. The $1 : l$ model on the left evaluates the length and the $1 : 1$ model on the right evaluates the strength. Note that the two traces in each model aim to different destination concepts, otherwise, it is covered by Case (I).

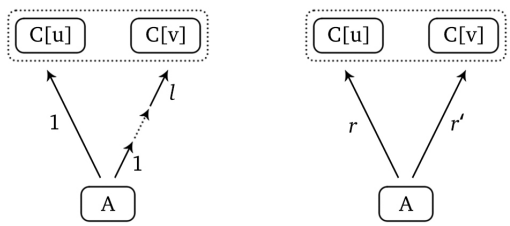

Figure 5: Models of Case (II). On the left is the $1 : l$ length model; On the right is the $1 : 1$ strength model.

3.2.3. Case (III): From Type II to Type I

Case (III) illustrates the opposite scenario of Case (II), where the source is a set containing two symmetrical concepts while the destination is a single concept, as demonstrated in Figure 6. As before, the $1 : l$ model on the left evaluates the granularity balance of the relationship by length and the $1 : 1$ model on the right evaluates that by strength. The symmetrical concepts pair $(C[u], C[v])$ forms a parallel concepts set, and the two traces from it to the

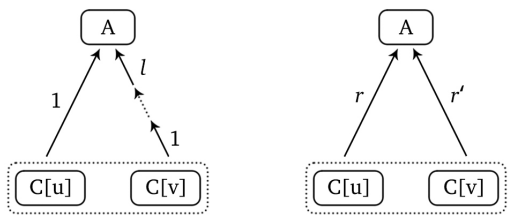

Figure 6: Models of Case (III). On the left is the $1 : l$ length model; On the right is the $1 : 1$ strength model.

single concept A should depart from different source concepts, otherwise, it is covered by Case (I).

3.2.4. Case (IV): From Type II to Type II

In Case (IV), the source and the destination of the traces in the models are both symmetrical concepts sets, as demonstrated in Figure 7. As the former cases, the $1 : l$ model on the left evaluates the granularity balance of the relationship by length and the $1 : 1$ model on the right evaluates that by strength. The pair $(C[u], C[v])$ represents two symmetrical concepts generated by the modifier pair (u, v) using context C, and $(C'[u'], C'[v'])$ represents two concepts generated by the modifier pair (u', v') using context C'. Note that the two traces in the models do not converge because otherwise the scenarios are covered by the former three cases.

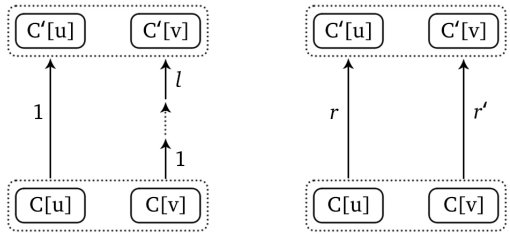

Figure 7: Models of Case (IV). On the left is the $1 : l$ length model; On the right is the $1 : 1$ strength model.

Recall the examples in Figure 2, we can discover that the example on the left of Figure 2 is an instance of Case (IV) 1:3 length model, and the example on the right is an instance of Case (I) strength model.

In short, we call the model which evaluates the granularity balance of relationship by length *the length model*, and correspondingly for *the strength model*. For a semantic relationship r with a hierarchy of subproperties such as Part-Of in FMA, when evaluating its granularity balance using length models, we consider all of the subproperties as r since every subproperty r' infers r.

3.3. Implementation

Leveraging the above models, we checked the granularity balances for IS-A and Part-Of relationships in two prominent biomedical ontologies: FMA and SNOMED CT. For SNOMED CT, we focused on the Body Structure subsets only. Since there is no subproperty for IS-A, only length models are needed to evaluate its granularity balance in both FMA and SNOMED CT. Besides, the Part-Of relationship is only evaluated in FMA because there does not exist Part-Of relationship in the Body Structure sub-hierarchy of SNOMED CT.

Preprocessing of Data: The OWL version v4.4.0 of FMA downloaded from its official site is stored in virtuoso and queried using SPARQL language, and SNOMED CT is stored in MySQL. Due to the large volume and complex structure of the two ontologies, only the $1 : 2$ and $1 : 3$ patterns for length models are considered in our experiments. Several datasets are prepared for future usage:

1. For the IS-A relationship: we compute a set $\mathbf{IR_1}$ which stores concept pairs related by one step of IS-A, a set $\mathbf{IR_2}$ which stores concept pairs indirectly related by two steps of IS-A's, and a set $\mathbf{IR_3}$ which stores concept pairs indirectly related by three steps of IS-A's.

2. For the Part-Of relationship: we compute a set $\mathbf{PR_1}$ which stores concept pairs related by one step of Part-Of, a set $\mathbf{PR_2}$ which stores concept pairs indirectly related by two steps of Part-Of's, and a set $\mathbf{PR_3}$ which stores concept pairs indirectly related by three steps of Part-Of's.
3. To retrieve Type II parallel concepts sets, i.e., symmetrical concepts sets, we compute a set $\mathbf{P}$ which stores eleven modifier pairs: At first, we retrieve all the modifier pairs in which the two members share some kind of common contexts in the concept names. Then we rank them by the frequencies of their appearances in the ontologies. The first ten pairs are:*(left, right), (anterior, posterior), (superior, inferior), (lateral,medial), (first, second), (second, third), (first, third), (fourth, third), (fourth, second), (lateral, anterior).* Together with *(upper, lower)*, the eleven modifier pairs comprise $\mathbf{P}$. In this study, the symmetrical concepts observed are restricted to those generated by them only.

Algorithm 1 Algorithm for computing Case (IV) length imbalances for Part-Of

Input: $\mathbf{PR_1}$; $\mathbf{PR_2}$; $\mathbf{PR_3}$; $\mathbf{P}$.

Output: Instances of $1:2$ and $1:3$ length imbalances for Case IV

```
for (A, B) ∈ R1 do
    for (u, v) ∈ P do
        if exists context C s.t. A == C[u] then A' = C[v]
        else if exists context C s.t. A == C[v] then A' = C[u]
        end if
        Collect all B' s.t. (A', B') ∈ R2, stored in S2
        for B' ∈ S2 do
            if B and B' are symmetrical then (A, B, A', B') is an instance of 1 : 2 imbalance
            end if
        end for
        Collect all B' s.t. (A', B') ∈ R3, stored in S3
        for B' ∈ S3 do
            if B and B' are symmetrical then (A, B, A', B') is an instance of 1 : 3 imbalance
            end if
        end for
    end for
end for
```

Algorithms: The computation procedures for preprocessing data are quite routine thus omitted in the paper. The main algorithms are divided into two groups: One group for computing length imbalances and the other group for computing strength imbalances. Algorithm 1 presents an algorithm in the first group: the procedure for computing instances of Case (IV) length imbalances for the Part-Of relationship. As explained

before, we do not differentiate the subproperties of Part-Of and consider them all Part-Of's. Algorithms for computing Case (I) to Case (III) length imbalance instances can be achieved easily by revising Algorithm 1. Procedures for calculating the imbalanced length instances for IS-A are similar.

Algorithm 2 presents an algorithm for computing Case (III) strength imbalances for Part-Of, a member in the second group. Procedures for computing strength imbalance instances of the other three cases can also be achieved easily by revising it. Note that the algorithms in the first group are applied on both FMA and SNOMED CT, while that in the second group are only needed in FMA.

Algorithm 2 Algorithm for computing Case (III) strength imbalances for Part-Of

Input: $\mathbf{PR_1}$; $\mathbf{P}$.

Output: Instances of $1:1$ strength imbalances for Case III

```
for (A, B) ∈ PR_1 and related by r do
    for (u, v) ∈ P do
        if exists context C s.t. A == C[u] then A' = C[v]
        else if exists context C s.t. A == C[v] then A' = C[u]
        end if
        if (A', B) ∈ PR_1 and related by r' and weight(r) ≠ weight(r') then
            (A, r, A', r', B) is an instance of length imbalance
        end if
    end for
end for
```

4. Results

After applying the preprocessing procedure on FMA, we achieved 104226 concept pairs directly related by one IS-A in $\mathbf{IR_1}$, 104064 concept pairs related by two-step IS-A's in $\mathbf{IR_2}$, and 103702 concept pairs related by three-step IS-A's in $\mathbf{IR_3}$. Those numbers for $\mathbf{PR_1}$, $\mathbf{PR_2}$ and $\mathbf{PR_3}$ in FMA are 61273, 82681 and 123893, respectively. On the other hand, the numbers of elements in $\mathbf{IR_1}$, $\mathbf{IR_2}$ and $\mathbf{IR_2}$ for IS-A in SNOMED CT's Body Structure subhierarchy are 38976, 64062, and 106769. They are all presented in Table 1 and Table 2.

4.1. Evaluation of IS-A in both FMA and SNOMED CT's Body Structure subhierarchy

The two groups of algorithms for discovering instances with imbalanced granularity are executed on the above concept pair sets. For the IS-A relationship, Table 1 illustrates the numbers of length-imbalanced instances of the four cases in FMA and SNOMED CT's Body Structure subhierarchy separately. For each case, the length imbalances are divided into two patterns: the 1:2 pattern and the 1:3 pattern. The number of instances for each pattern is presented individually. As shown in Table 1, more than 85% of the instances

take the 1:2 pattern, in both FMA and SNOMED CT, and more than 50% of them are Case (III) instances. On the other hand, it is remarkable that totally there are only more than 100 length-imbalanced instances in both ontologies, considering the vast amount of concept pairs in $\mathbf{IR_1}$, $\mathbf{IR_2}$ and $\mathbf{IR_3}$.

Table 1: Results for IS-A in FMA and Snomed CT's Body Structure subset

Ontologies	**Pair Counts**			**Case I**		**Case II**		**Case III**		**Case IV**		**Total**	
	$\mathbf{IR_1}$	$\mathbf{IR_2}$	$\mathbf{IR_3}$	1:2	1:3	1:2	1:3	1:2	1:3	1:2	1:3	1:2	1:3
FMA	104226	104064	103702	15	3	2	-	61	11	29	-	(107, 88%)	(14, 12%)
SNOMED CT	38976	64062	106769	1	-	5	1	75	11	24	7	(105, 85%)	(19, 15%)

Figure 8 demonstrates four examples with imbalanced length for the IS-A relationship in the two terminologies. To save space, only one example is presented for each case. Note that there are two duplicate nodes *Lateral cord of brachial nerve plexus* in Ex.1, and the 1:2 length imbalance is actually caused by the triple *(Lateral cord of brachial nerve plexus, IS-A, Anterior cord of brachial nerve plexus)*, which is not correct.

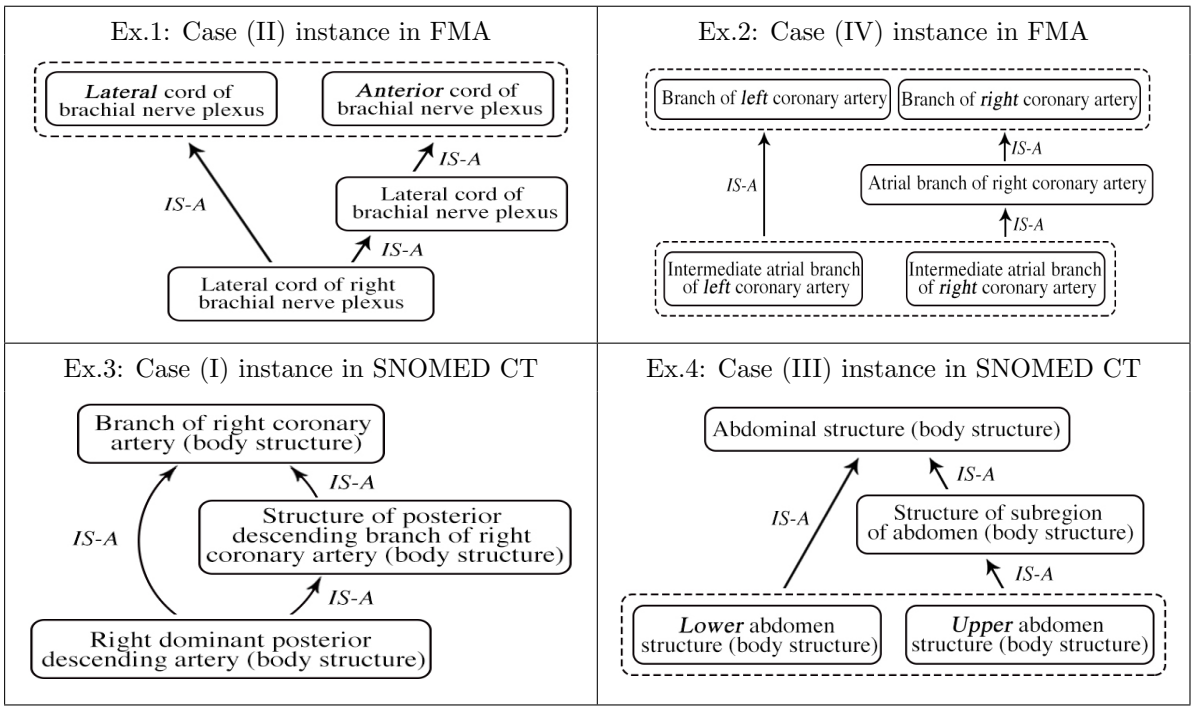

Figure 8: Length-imbalanced instances for the IS-A relationship: the upper two examples are in FMA, and the lower two examples are in SNOMED CT.

4.2. Evaluation of Part-Of in FMA

Table 2 presents the results of evaluating the granularity balance of the Part-Of relationship in FMA using the length models and the strength models of the four cases (see Figure 4 to Figure 7). The above part

Table 2: Results for Part-Of in FMA

Models	**Pair Counts**			**Case I**		**Case II**		**Case III**		**Case IV**		**Total**	
	$\mathbf{PR_1}$	$\mathbf{PR_2}$	$\mathbf{PR_3}$	1:2	1:3	1:2	1:3	1:2	1:3	1:2	1:3	1:2	1:3
Length	61,273	82,681	123,893	1,807	513	107	123	1,000	208	1,048	129	3,962	973
Strength	regional-part-of vs. branch-of			850		3		319		502		(1674, 89.7%)	
	regional-part-of vs. tributary-of			72		5		64		36		(177, 9.5%)	
	part-of vs. regional-part-of			-		-		3		4		(7, 0.37%)	
	part-of vs. constitutional-part-of			4		-		-		4		(8, 0.43%)	

illustrates the distribution of the length-imbalanced instances on the 1:2 and the 1:3 patterns from Case (I) to Case (IV). As it shows, 3962 out of the total 4935(80%) instances are 1:2 length imbalances. Among all the four cases, Case (I) turns out to be the most common one (almost 50%), and Case (II) has the least number of instances.

The bottom part of Table 2 shows the distribution of the strength-imbalanced instances on different pairs of subproperties of the Part-Of relationship in the four cases. The percentages in the last column are calculated based on the total number of strength-imbalanced instances. The Part-Of relationship in FMA has six subproperties(including itself) on three levels(see Figure 1), while results in Table 2 illustrate that the strength imbalances only take place in the following four pairs: *regional-part-of vs.branch-of, regional-part-of vs.tributary-of, part-of vs.regional-part-of* and *part-of vs.constitutional-part-of,* and nearly 90% of them occur between *regional-part-of* and *branch-of.* Besides, more than a half of all the instances with imbalanced strength are Case (I) imbalances, while Case (II) occupies only a negligible portion.

Figure 9 demonstrates four examples with imbalanced granularity of the Part-Of relationship in FMA, including two length imbalances and two strength imbalances.

4.3. Distribution on modifiers

Except for Case (I), the other three cases all have Type II parallel concepts involved in their models. Since the Type II parallel concepts considered in this study are symmetrical concepts constructed by the eleven symmetrical modifier pairs in **P** with some contexts, for those parallel concepts involved in our imbalanced instances, we performed an analysis of their distribution on the fourteen modifiers appeared in the eleven symmetrical modifier pairs. The results are presented in Table 3. To sketch a clear picture, the fourteen modifiers are shuffled into five modifier groups in Table 3 according to their anatomical positions. The percentages are counted on the total number of each column. Note that every instance of Case (IV) will be counted twice since it includes two pairs of symmetrical concepts. It is detectable from Table 3 that the results for SNOMED CT are quite even on the five groups. In FMA, the imbalances for IS-A happen

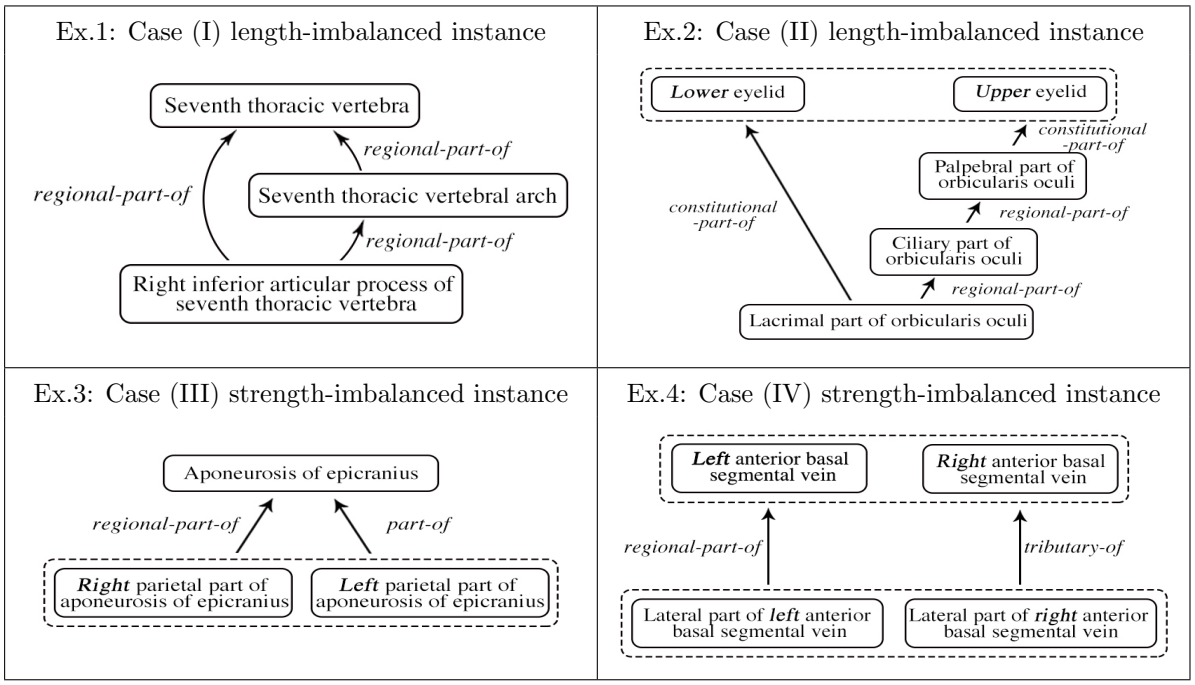

Figure 9: Instances with imbalanced granularity for the Part-Of relationship: the upper two examples are length imbalances, and the lower two examples are strength imbalances.

most frequently in the group {*anterior,medial,lateral,posterior*}, and that for Part-Of focus on the pair *(left, right)*.

Table 3: Distribution of Type II parallel concepts on the modifiers

Modifier Groups	**IS-A**				**Part-Of**			
	FMA		**SNOMED CT**		length	strength	total	
{left,right}	16	12%	30	19%	2220	1048	3268	62%
{anterior,medial,lateral,posterior}	59	45%	32	21%	462	187	649	12%
{superior,inferior}	18	14%	28	18%	147	73	220	4%
{first,second,third,fourth}	35	26%	36	24%	894	176	1070	20%
{upper,lower}	4	3%	28	18%	69	2	71	1%

To summarize, for length models, over 80% of the instances are 1:2 length imbalances for both IS-A and Part-Of, regardless of the ontology. For strength models, experiment on the Part-Of relationship in FMA indicates that 90% of the strength imbalances occur between *regional-part-of* and *branch-of*. As for distribution on the four cases, the most common case for IS-A is Case (III), while that for Part-Of is Case (I).

4.4. Result analysis

As we can see, the instances of our evaluation models may not suffice to be errors, however, they can be improved to some extent. We classify the instances with imbalanced granularity of relationships into the following three groups:

1. **Redundancies**. All instances of Case I(see Figure 4) are redundancies. About the $1 : l$ length model, due to the transitive features of IS-A and Part-Of, the direct trace with length 1 is redundant. For instance, in Ex.3 presented in Figure 8, the left direct IS-A edge is redundant. About the strength model, suppose r' is a subproperty of r, then (C, r', P) infers (C, r, P), thus the (C, r, P) trace is redundant. The example on the right of Figure 2 illustrates an instance of this situation: Since *branch-of* is a subproperty of *regional-part-of* (see Figure 1), the edge *regional-part-of* is redundant.
2. **Inconsistencies**. Strength models of Case (II), Case (III) and Case (IV)(see Figure 5 to Figure 7) indicate inconsistencies. An instance of Case (III) is demonstrated by Ex.3 in Figure 9: Although *regional-part-of* infers *part-of*, the reverse does not hold true, which means, the existence of *(Left parietal part of aponeurosis of epicranius, part-of, Aponeurosis of epicranius)* does not necessarily indicate the existence of *(Left parietal part of aponeurosis of epicranius, regional-part-of, Aponeurosis of epicranius)*, thus introduces inconsistency. A solution for this problem is replacing the subproperty *regional-part-of* with *part-of*, which keeps coherent with the original information.
3. **Revise suggestions**. Except for Case (I), the length models of the other three cases suggest the possibility of introducing new concepts and relations to the ontology, or deleting existing concepts and relations if the granularity is supposed to be coarse. For instance, the example on the left of Figure 2 provides two suggestions: adding the two concepts *Structure of soft tissues of lower extremity* and *Skin and subcutaneous tissue structure of lower limb* to the left trace, or deleting their symmetrical concepts from the existing structure. Usually, the former suggestion is more feasible, so instances in this group can also be viewed as cases of missing concepts, or missing relations if the concepts exist in the ontology already, or redundancies if both the concepts and the relations exist.

According to the above classification, we rearranged the results in Table 4. The percentages are counted on the total number of each column. As shown by Table 4, most of the instances for IS-A suggest revisions, which is reasonable because there is no strength imbalances for it. Moreover, 48% of the instances for Part-Of are redundancies because Case (I) is the most common case for it.

One thing needed to be pointed out is that there may be instances with mixed cases, so the results in the four cases are not necessarily disjoint if only concept nodes are concerned. For instance, the whole

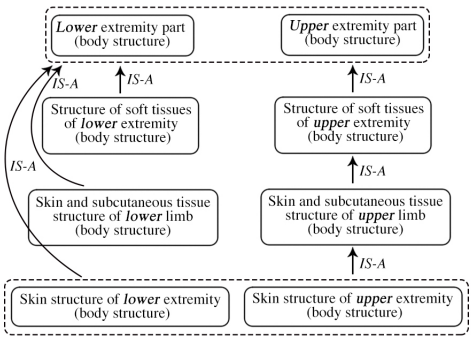

Figure 10: A mixed-case instance in SNOMED CT

Table 4: Classifying imbalanced instances into three groups

Classification	**FMA**				**SNOMED CT**	
	IS-A		**Part-Of**		**IS-A**	
Redundancy	18	15%	3,246	48%	1	1%
Inconsistency	-	-	940	14%	-	-
Revise Suggestion	103	85%	2,615	38%	123	99%

picture for the left example in Figure 2 is depicted in Figure 10, which also contains a Case (IV) 1:2 length model for the pair *(Skin and subcutaneous tissue structure of lower limb, Skin and subcutaneous tissue structure of upper limb)*. Apparently, to amend this example, we need to add two IS-A relations to the left side and then delete two redundant ones. There are many other mix-case instances in the ontologies. No matter what, for parallel concepts sets introduced in this study, all instances with imbalanced granularity of relationship are within the scope of and will be captured by the four cases we extracted.

5. Discussions

Given that different biomedical ontologies are designed under different circumstances and used for different scenarios, it is reasonable for them to behave differently, thus have distinct granularity balances for the same semantic relationship. However, within the scope of a particular ontology, the granularity of a relationship is supposed to be balanced. We believe that better inner characteristics of ontology is essential for better ontology enhancement and integration.

The approach for examining granularity balances of semantic relationships proposed in this paper relies on the discovery of parallel concepts sets. A single concept is surely parallel to itself, but how to find more parallel concepts? It depends on the unique feature of each ontology and the knowledge field it describes. For instance, based on the largely bilaterally symmetrical feature of human anatomy system, we found out that symmetrical concepts in FMA and SNOMED CT's Body Structure subsets suffice to be parallel concepts in this study. To discover more parallel concepts, human efforts may be involved when necessary. Another question arises is that do we really need parallel concepts sets with more than two elements. Since granularity imbalances are discovered when the traces between parallel concepts sets are imbalanced, and each trace has only one source and one destination, essentially the granularity imbalances happen between pairs of traces. As a result, parallel concepts sets with two elements are enough for uncovering granularity imbalances.

There are two reasons for not combining the length model and the strength model together: One is that they discover different aspects of granularity imbalance, including redundancies, inconsistencies and revise

suggestions, as demonstrated in the results. The other reason is that the semantic weight assigned to a relationship may prevent the reveal of length imbalance. For instance, suppose there is a subproperty r_1 with semantic weight k and another subproperty r_2 with semantic weight 1, then the semantic distance of one r_1 will be equal to that of k r_2's, which prevents the detection of $1 : k$ length imbalance.

Our work is close to that of Zhe He [27, 28] but differs with it in three aspects: Firstly, their study focuses on differentiation across terminologies while we pay attention to the intrinsic imbalances of semantic relationships inside terminologies. Secondly, the set of topological patterns proposed in [28] only illustrate cases of length imbalances but no strength imbalances, because they only consider the IS-A relationship. Lastly, the structurally congruent concepts in [27, 28] were defined between two single concepts while ours are defined between two sets of parallel concepts, which shows that our model is more general.

Limitation. Although the models presented in this study are general and comprehensive, taking the large volume of ontologies and the complex structures of the Part-Of relationship into account, our algorithms only focus on the 1:2 and the 1:3 patterns for length models. Although our results demonstrate that most of the instances take the 1:2 pattern, to obtain exhaustive results for other patterns, big data approaches such as MapReduce cloud computing will be needed to improve algorithm efficiency [23].

6. Conclusion

In this study, we proposed a systematic mechanism for evaluating the granularity balances of semantic relationships within large biomedical ontologies for quality assurance. Applications of the evaluation models on the IS-A and Part-Of relationships were conducted in FMA and SNOMED CT. A considerable amount of granularity imbalances were uncovered, including redundancies, inconsistencies and potential suggestions for importing new concepts into the ontology or deleting existing concepts from the ontology according to different relationship granularity levels requested by the ontology.

Our methodology is innovative in three aspects: First, to our knowledge, it is the first time for us to pay attention to the inner granularity balances of relationships within biomedical ontologies for quality assurance. Second, we defined parallel concepts and utilized them to design the general evaluation models, which provide a scalable approach easily adoptable by other semantic relationships for studying their granularity. Third, the Part-Of relationship in FMA was less frequently audited than the IS-A relationship in the literature due to its complex subhierarchy, while the comprehensive analysis of it performed in this study will make it easier to handle for future auditing tasks.

7. Acknowledgements

This research is supported by the following grants:*National Science Foundation of China(No. 61502221, No. 61402220)*, *Scientific Research Fund of Hunan Provincial Education Department for Excellent Tal-*

ents (No. 14B153, No.16C1378), the Philosophy and Social Science Foundation of Hunan Province (No. 14YBA335) .

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